

Linear Programming

LP Geometry

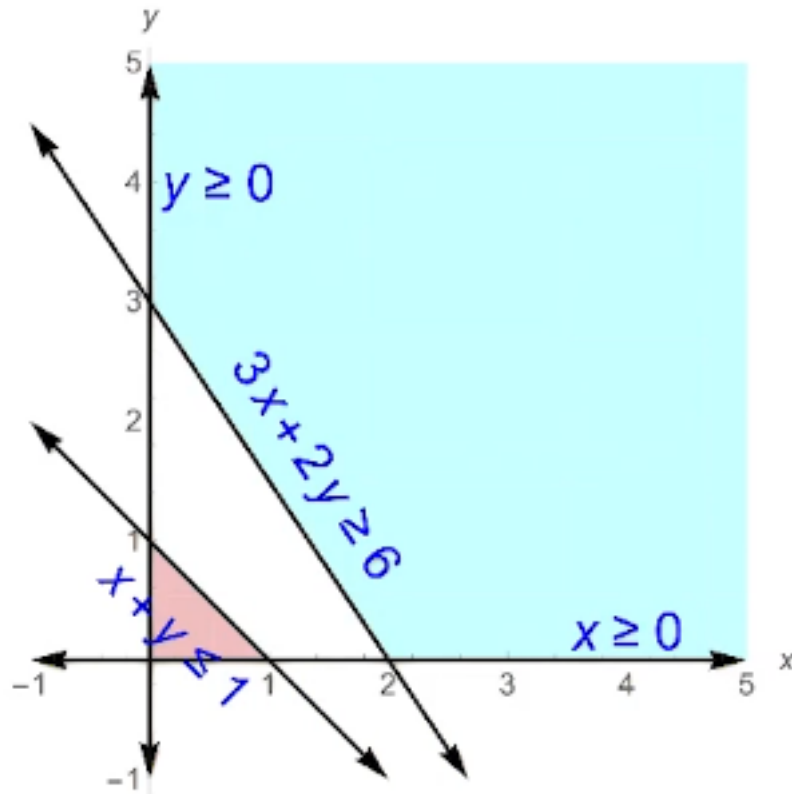
1. Consider a linear program in standard form
 - n variables
 - m constraints
 - Feasible region = convex set
 - Simplex algorithm walks on vertices

LP Optimum

1. Optimum of LP is achieved at a vertex of the feasible region except if:
 - Infeasible: Feasible region is empty
 - Objective:
 - * $\max(5x-7y)$
 - Constraints:
 - * $x + y \leq 1$
 - * $3x + 2y \geq 6$
 - * $x, y \geq 0$
 - Unbounded

Infeasible Example

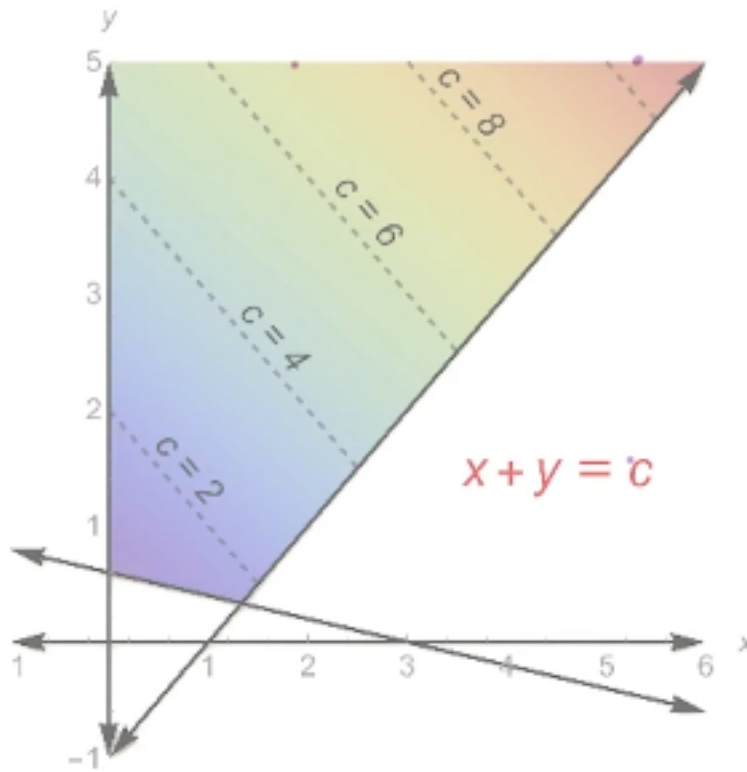
1. Feasible region is empty
 - This is regardless of the objective function



Infeasible Region

Unbounded Example

1. Unbounded: Optimal is arbitrarily large
 - Objective:
 - $\max(x+y)$
 - Constraints:
 - $x - y \leq 1$
 - $x + 5y \geq 3$
 - $x, y \geq 0$



Unbounded Region

Unbounded to Bounded

- Whether an LP is bounded or unbounded depends on the objective function and constraints
 - Objective:
 - $\max(2x-3y)$
 - Constraints:
 - $x - y \leq 1$
 - $x + 5y \geq 3$
 - $x, y \geq 0$
- Whether an LP is feasible or infeasible depends on the constraints

LP Optimum - Part 2

- To determine if a feasible LP is bounded, we must consider the dual of the LP

Infeasible?

- Is there any x satisfying?
 - Make a new variable z
 - z can be positive or negative, while x must be greater than zero
 - Add the new variable z to the constraints
 - This allows us to make the constraint trivially satisfiable
 - Run the LP and maximize z
 - If there's a solution where z is greater than 0, the LP is feasible

- This provides a feasible point for the LP if it is feasible
 - * Starting point for simplex algorithm
- Objective:
 - $\max(z)$
- Constraint:
 - $Ax + z \leq b$
 - $x \geq 0$